



NORTH SYDNEY BOYS HIGH SCHOOL

MATHEMATICS (EXTENSION 1)

2017 HSC Course Assessment Task 4

Friday August 18, 2017

General instructions

- Working time – 60 minutes.
(plus 5 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets used in the correct order within this paper and hand to examination supervisors.
- Commence each new question on a new page. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

STUDENT NUMBER: **# BOOKLETS USED:**

Class (please ✓)

☐ 12M3A – Ms Ziaziaris

☐ 12M4A – Dr Jomaa

☐ 12M3B – Mr Berry

☐ 12M4B – Miss Lee

☐ 12M3C – Mr Hwang

☐ 12M4C – Mr Lin

☐ 12M3D – Mr Ireland

Marker's use only.

QUESTION	1	2	3	4	Total	%
MARKS	$\overline{10}$	$\overline{8}$	$\overline{11}$	$\overline{5}$	$\overline{34}$	

Question 1 (10 Marks)

Commence a NEW page.

Marks

When a particle moves along the x -axis with velocity v at time t , its acceleration is given by

$$\ddot{x} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right).$$

The velocity $v \text{ ms}^{-1}$ of a particle moving in simple harmonic motion along the x -axis is given by the expression

$$v^2 = 27 + 18x - 9x^2.$$

- | | | |
|-----|--|----------|
| (a) | Between which two points is the particle oscillating? | 2 |
| (b) | What is the amplitude of the motion? | 1 |
| (c) | Find the acceleration in terms of x . | 2 |
| (d) | What is the centre of the motion? | 1 |
| (e) | Find the period of the oscillation. | 2 |
| (f) | If the particle starts from the position furthest to the right, find the displacement function in terms of t . | 2 |

Question 2 (8 Marks)

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Marks

A Projectile is fired from O with velocity V at an angle of inclination θ across level ground. The equations of the motion of the projectile are

$$x = Vt \cos \theta \quad \text{and} \quad y = Vt \sin \theta - \frac{1}{2}gt^2 \text{ (Do Not Prove this.)}$$

- | | | |
|-----|---|----------|
| (a) | Show that the Cartesian equation of the path can be written as | 2 |
| | $y = \frac{-gx^2}{2V^2} \sec^2 \theta + x \tan \theta.$ | |
| (b) | Show that the Cartesian equation of the path can be written as | 2 |
| | $gx^2 \tan^2 \theta - 2xV^2 \tan \theta + (2yV^2 + gx^2) = 0.$ | |
| (c) | Suppose that $V = 200 \text{ ms}^{-1}$, $g = 10 \text{ ms}^{-2}$ and the projectile hits a target positioned 3000 m horizontally and 500 m vertically from the origin. Show that | 2 |
| | $\tan \theta = \frac{4 \pm \sqrt{3}}{3}.$ | |
| (d) | Hence find the two possible values of θ correct to the nearest minute. | 2 |

Please Turn Over

Question 3 (11 Marks)

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Marks

Assume that the tides rise and fall in simple harmonic motion. At low tide the channel in a harbour is 8 m deep and at high tide it is 12 m deep. Low tide is at 9 a.m. and high tide is at 4 p.m.

- | | | |
|-----|--|----------|
| (a) | What is the depth of the channel at $12 : 30\text{ p.m.}$? | 1 |
| (b) | What is the amplitude of the motion? | 1 |
| (c) | What is the period of the motion? | 1 |
| (d) | Show that the depth $y\text{ m}$ of the water in the channel is given by $y = 10 - 2 \cos\left(\frac{\pi}{7}t\right)$, where t is the number of hours after low tide. | 2 |
| (e) | Draw the graph of $y = 10 - 2 \cos\left(\frac{\pi}{7}t\right)$, for $0 \leq t \leq 14$. | 2 |
| (f) | A ship needs at least 9 m of water to pass through the channel safely. Mark on your graph the times when the ship can safely navigate the channel. | 2 |
| (g) | Algebraically, find the times the ship can navigate safely through the channel. | 2 |

Please Turn Over

Question 4 (5 Marks)

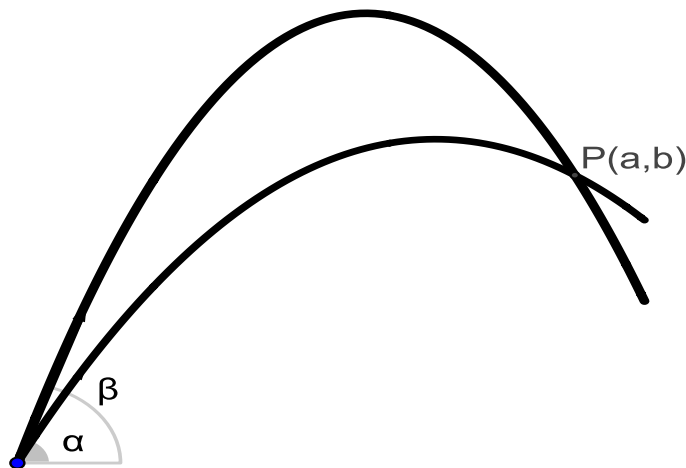
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Marks

The equations of the motion of the projectile are

$$x = vt \cos \theta, \quad y = vt \sin \theta - \frac{1}{2}gt^2 \quad (\text{Do Not Prove This}).$$

Two particles are projected simultaneously from O in different angles of inclination with the same speed V so as to pass through another point $P(a, b)$. If α, β are the angles of projection.



- (a) Prove that they pass through P at time separated by

3

$$t_1 - t_2 = 2 \frac{V}{g} \frac{\sin(\alpha - \beta)}{\cos \alpha + \cos \beta}.$$

- (b) Given that

2

$$\cos A + \cos B = 2 \cos \left(\frac{A+B}{2} \right) \cos \left(\frac{A-B}{2} \right).$$

Show that

$$t_1 - t_2 = 2 \frac{V}{g} \sin \left(\frac{\alpha - \beta}{2} \right) \sec \left(\frac{\alpha + \beta}{2} \right).$$

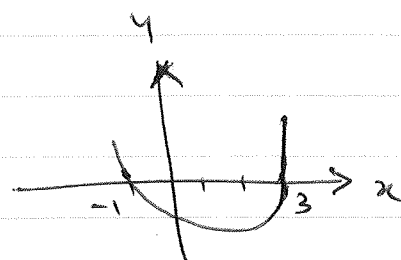
End of paper.

Question 1:

$$v^2 = 27 + 18x - 9x^2$$

$$\begin{aligned} \text{(a)} \quad v^2 &= -9(x^2 - 2x - 3) \\ &= -9(x-3)(x+1) \\ v^2 \geq 0 &\therefore (x-3)(x+1) \leq 0 \end{aligned}$$

$$\therefore -1 \leq x \leq 3$$



The particle is oscillating between $x = -1$ and $x = 3$.

$$\text{(b)} \quad \text{Amplitude} = \frac{3 - (-1)}{2} = 2$$

$$\begin{aligned} \text{(c)} \quad \ddot{x} &= \frac{d}{dx} \left(\frac{1}{2} v^2 \right) \\ &= \frac{d}{dx} \left[\frac{1}{2} (27 + 18x - 9x^2) \right] \\ &= \frac{1}{2} (18 - 18x) \\ &= -9(x-1) \end{aligned}$$

(d) The centre of the motion is $x = 1$

$$\text{(e)} \quad \text{from (c)} \quad n = 3, \therefore T = \frac{2\pi}{n} = \frac{2\pi}{3}$$

$$\text{(f)} \quad x = 1 + 2 \cos(3t + \alpha)$$

$$\text{At } t=0, x=3$$

$$\therefore 3 = 1 + 2 \cos(\alpha) \therefore \cos \alpha = 1 \therefore \alpha = 0$$

$$\boxed{x = 1 + 2 \cos 3t}$$

Question 2:

$$x = vt \cos \theta, \quad y = vt \sin \theta - \frac{1}{2}gt^2$$

$$(a) \quad t = \frac{x}{v \cos \theta}$$

$$\therefore y = v \sin \theta \times \frac{x}{v \cos \theta} - \frac{1}{2}g \left(\frac{x}{v \cos \theta} \right)^2$$

$$= \frac{-gx^2}{2v^2 \cos^2 \theta} + \tan \theta x$$

$$= \frac{-gx^2}{2v^2} \sec^2 \theta + x \tan \theta \quad \left(\sec^2 \theta = \frac{1}{\cos^2 \theta} \right)$$

$$(b) \quad \sec^2 \theta = 1 + \tan^2 \theta$$

$$\text{from (a)} \quad 2v^2 y = -gx^2 \sec^2 \theta + 2v^2 x \tan \theta$$

$$gx^2 (1 + \tan^2 \theta) - 2v^2 x \tan \theta + 2v^2 y = 0$$

$$gx^2 \tan^2 \theta - 2v^2 x \tan \theta + (2yv^2 + gx^2) = 0 //$$

$$(c) \quad v = 200 \text{ m/s}, \quad g = 10 \text{ m/s}^2, \quad x = 3 \text{ km} = 3000 \text{ m}, \quad y = 0.5 \text{ km} = 500 \text{ m}$$

From (b)

$$10 \times (3000)^2 \tan^2 \theta - 2 \times (200)^2 \times 3000 \tan \theta + 2 \times 500 \times (200)^2 + 10 \times (3000)^2 = 0$$

$$9 \times 10^7 \tan^2 \theta - 24 \times 10^7 \tan \theta + 4 \times 10^7 + 9 \times 10^7 = 0$$

$$10^7 (9 \tan^2 \theta - 24 \tan \theta + 13) = 0$$

$$9 \tan^2 \theta - 24 \tan \theta + 13 = 0$$

$$\Delta = 24^2 - 4 \times 9 \times 13 = 108 = 3 \times 6^2$$

$$\sqrt{\Delta} = 6\sqrt{3}$$

$$\tan \theta = \frac{24 \pm 6\sqrt{3}}{2 \times 9} = \frac{4 \pm \sqrt{3}}{3} //$$

d

from (c)

$$\tan \theta = \frac{4+\sqrt{3}}{3} \quad \therefore \theta = 62^{\circ}22'$$

or

$$\tan \theta = \frac{4-\sqrt{3}}{3} \quad \therefore \theta = 37^{\circ}5'$$

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Question 3:

low tide at 9:00 a.m. $y = 8\text{ m}$

High tide at 4:00 p.m. $y = 12\text{ m}$

(a) At 12:30 p.m., the depth of the channel
$$= \frac{8+12}{2} = 10\text{ m}$$

(b) Amplitude $= \frac{12-8}{2} = 2$

(c) let $t=0$ at 9:00 a.m. $\therefore t=7$ at 4:00 p.m.
 $\therefore$ period of the motion is 14 hours.

(d) $y = 10 - 2 \cos\left(\frac{\pi}{7}\right)t$

$$\text{period} = 14 = \frac{2\pi}{n} \therefore n = \frac{2\pi}{14} = \frac{\pi}{7}$$

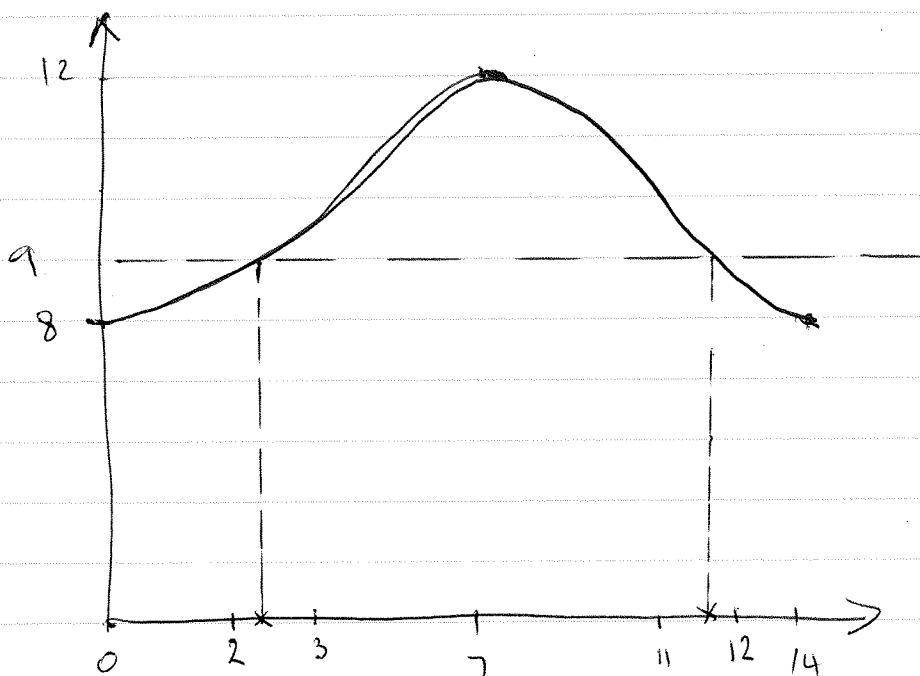
At $t=0$, $y = 10 - 2 \cos 0 = 10 - 2 = 8$ (9:00 a.m.)

At $t=7$, $y = 10 - 2 \cos \pi = 10 + 2 = 12$ (4:00 p.m.)

At $t=3.5$, $y = 10 - 2 \cos \frac{\pi}{2} = 10$ (12:30 p.m.)

(e)

(f)



(9)

$$9 = 10 - 2 \cos\left(\frac{\pi}{7}t\right)$$

$$2 \cos\left(\frac{\pi}{7}t\right) = 1 \quad \therefore \cos\left(\frac{\pi}{7}t\right) = \frac{1}{2}$$

$$\frac{\pi}{7}t = \pi/3 \quad \text{or} \quad 5\pi/3$$

$$t = 7/3 \quad \text{or} \quad 35/3$$

$$= 2\frac{1}{3} \quad \text{or} \quad 11\frac{2}{3}$$

2 hours and 20 min or 11 hours and 40 min

Between 9: am + 2:20 = 11:20 a.m.

9: am + 11:40 = 8:40 p.m. //

Question 4: $x = vt \cos \theta$, $y = vt \sin \theta - \frac{1}{2} g t^2$

(a) $a = vt_1 \cos \alpha$, $b = vt_1 \sin \alpha - \frac{1}{2} g t_1^2$
 $a = vt_2 \cos \beta$, $b = vt_2 \sin \beta - \frac{1}{2} g t_2^2$

$$vt_1 \sin \alpha - \frac{1}{2} g t_1^2 = vt_2 \sin \beta - \frac{1}{2} g t_2^2$$

$$vt_1 \sin \alpha - vt_2 \sin \beta - \frac{1}{2} g (t_1^2 - t_2^2) = 0$$

$$t_1 = \frac{a}{v \cos \alpha}, \quad t_2 = \frac{a}{v \cos \beta}$$

$$\frac{a v \sin \alpha}{v \cos \alpha} - \frac{a v \sin \beta}{v \cos \beta} - \frac{1}{2} g (t_1 - t_2)(t_1 + t_2) = 0$$

$$\boxed{t_1 + t_2 = \frac{a}{v} \left(\frac{1}{\cos \alpha} + \frac{1}{\cos \beta} \right)}$$

$$a \left(\frac{\sin \alpha \cos \beta - \cos \alpha \sin \beta}{\cos \alpha \cos \beta} \right) - \frac{1}{2} g \times \frac{a}{v} \left(\frac{1}{\cos \alpha} + \frac{1}{\cos \beta} \right) (t_1 - t_2) = 0$$

$$\frac{\sin(\alpha - \beta)}{\cos \alpha \cos \beta} - \frac{g}{2v} \frac{(\cos \alpha + \cos \beta)}{\cos \alpha \cos \beta} (t_1 - t_2) = 0$$

$$t_1 - t_2 = \frac{2v}{g} \frac{\sin(\alpha - \beta)}{\cos \alpha + \cos \beta}$$

///

(b)

$$\cos \alpha + \cos \beta = 2 \cos \left(\frac{\alpha + \beta}{2} \right) \cos \left(\frac{\alpha - \beta}{2} \right)$$

$$\sin(\alpha - \beta) = 2 \sin \left(\frac{\alpha - \beta}{2} \right) \cos \left(\frac{\alpha - \beta}{2} \right)$$

$$t_1 - t_2 = \frac{2v}{g} \times \frac{\cancel{2} \sin \left(\frac{\alpha - \beta}{2} \right) \cancel{\cos \left(\frac{\alpha - \beta}{2} \right)}}{\cancel{2} \cos \left(\frac{\alpha + \beta}{2} \right) \cancel{\cos \left(\frac{\alpha - \beta}{2} \right)}}$$

$$= \frac{2v}{g} \sin \left(\frac{\alpha - \beta}{2} \right) \sec \left(\frac{\alpha + \beta}{2} \right).$$

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